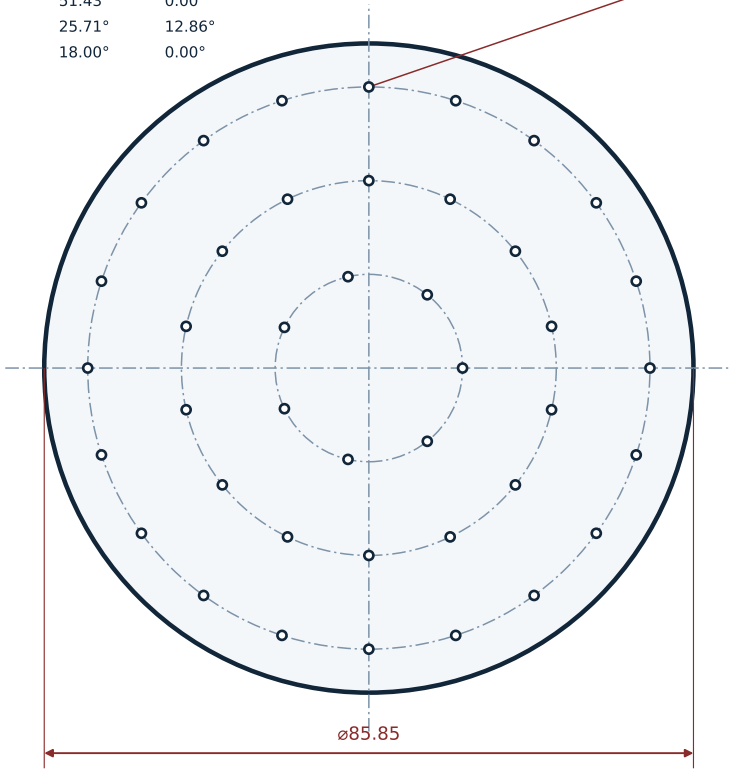


RING DATA

RING	B.C. ø	HOLES	PITCH	STAGGER
1	24.78	7	51.43°	0.00°
2	49.57	14	25.71°	12.86°
3	74.35	20	18.00°	0.00°

41× ø1.200 THRU
L/D 10.58 AREA 46.37 mm²



VIEW ON INLET (UPSTREAM) FACE

EDGE VIEW — HOLES PROJECTED, ALL THRU



12.70 THK

HOLE TABLE

mm from plate centre, X right / Y up

NO	RING	X	Y	R	ANG
1	1	+12.392	+0.000	12.392	0.00°
2	1	+7.726	+9.688	12.392	51.43°
3	1	-2.757	+12.081	12.392	102.86°
4	1	-11.164	+5.377	12.392	154.29°
5	1	-11.164	-5.377	12.392	205.71°
6	1	-2.757	-12.081	12.392	257.14°
7	1	+7.726	-9.688	12.392	308.57°
8	2	+24.162	+5.515	24.783	12.86°
9	2	+19.376	+15.452	24.783	38.57°
10	2	+10.753	+22.329	24.783	64.29°
11	2	+0.000	+24.783	24.783	90.00°
12	2	-10.753	+22.329	24.783	115.71°
13	2	-19.376	+15.452	24.783	141.43°
14	2	-24.162	+5.515	24.783	167.14°
15	2	-24.162	-5.515	24.783	192.86°
16	2	-19.376	-15.452	24.783	218.57°
17	2	-10.753	-22.329	24.783	244.29°
18	2	-0.000	-24.783	24.783	270.00°
19	2	+10.753	-22.329	24.783	295.71°
20	2	+19.376	-15.452	24.783	321.43°
21	2	+24.162	-5.515	24.783	347.14°
22	3	+37.175	+0.000	37.175	0.00°
23	3	+35.355	+11.488	37.175	18.00°
24	3	+30.075	+21.851	37.175	36.00°
25	3	+21.851	+30.075	37.175	54.00°
26	3	+11.488	+35.355	37.175	72.00°
27	3	+0.000	+37.175	37.175	90.00°
28	3	-11.488	+35.355	37.175	108.00°
29	3	-21.851	+30.075	37.175	126.00°
30	3	-30.075	+21.851	37.175	144.00°
31	3	-35.355	+11.488	37.175	162.00°
32	3	-37.175	+0.000	37.175	180.00°
33	3	-35.355	-11.488	37.175	198.00°
34	3	-30.075	-21.851	37.175	216.00°
35	3	-21.851	-30.075	37.175	234.00°
36	3	-11.488	-35.355	37.175	252.00°
37	3	-0.000	-37.175	37.175	270.00°

... 4 more — export the hole table CSV

NOTES

- ALL DIMENSIONS IN MILLIMETRES. ORIGIN AT PLATE CENTRE. PLAN VIEW IS FROM THE INLET (UPSTREAM) FACE.
- 41 HOLES ø1.200 THRU, EQUALLY SPACED ON THE BOLT CIRCLES LISTED IN RING DATA.
- FLOW AREA GOES AS d²: A HOLE ø ERROR OF +0.02 GIVES +3.4% AREA, AND THE SAME SHIFT IN OXIDISER FLOW. SET THE TOLERANCE TO SUIT.
- Cd 0.700 ASSUMED FOR SHARP-EDGED DRILLED HOLES. DEBURR THE OUTLET FACE. DO NOT CHAMFER OR RADIUS THE INLET WITHOUT RE-SIZING — INLET GEOMETRY CHANGES Cd.
- L/D 10.58 FROM 12.70 THICKNESS. THE FLOW MODEL ASSUMES THIS LENGTH; CHANGING THICKNESS CHANGES PREDICTED FLOW.
- MIN WEB BETWEEN HOLES 9.55. EDGE MARGIN 5.15.
- PRINT AT 100% ON A4 LANDSCAPE FOR TRUE SCALE — CONFIRM WITH THE CHECK BAR BEFORE MEASURING OFF THIS SHEET.
- SIZED FOR 1.0 kg/s N2O. SEE TRADE STUDY FOR THE BASIS OF THIS CHOICE.

CHECK BAR — 50 mm ON PAPER:



INJECTOR ORIFICE PLATE

PROJECT	Mini Hybrid V2.2
PART NO	INJ-PLATE-V2.2B
MATERIAL	—
PLATE	ø85.85 × 12.70 THK
HOLES	41 × ø1.200 THRU
SCALE	1 : 1 UNITS mm
DRAWN	Molen 2026-09-21
HASTE v1.0.0 — HEM model, Cd 0.700	